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‘AURORA POD’ PROPOSAL

Aurora – Accessible Embedded Systems Sleep Pod

Aurora is an accessibility-first embedded systems project exploring how to design a personal sleep-pod environment around the needs of disabled students in university accommodation.

This project has five main objectives:

- Design an accessible sleep-pod concept
- Explore environmental monitoring
- Prototype one or more embedded control features
- Record and analyse useful sensor data
- Document design decisions and testing

Additionally, the design must be functional and practically achievable for a university student to create. Some of the design constraints involve:

- Must fit in a university room with a footprint no larger than a standard double bed
- There must be no permanent wall or ceiling attachments
- The design should minimise unnecessary material use and environmental impact where practical
- Accessibility must be a main priority for all features and designs
- It must be achievable within a student budget
- It must be safe and removable

Accessibility Considerations

Because accessibility is the central focus of the Aurora Pod, the design will consider a broad range of functional needs rather than being designed around specific medical conditions or diagnoses.

The main accessibility considerations are:

Mobility and Physical Access

- Limited mobility or range of movement
- Reduced strength or endurance
- Limited dexterity or fine motor control
- Different reach ranges and movement patterns
- Use of mobility or accessibility equipment

Sensory Access

- Low vision or blindness
- Deafness or reduced hearing
- Sensitivity to light, sound or other sensory stimulation
- Difficulty distinguishing information presented using colour alone

Cognitive and Interface Accessibility

- Difficulty processing complex interfaces
- Fatigue or reduced concentration
- Preference for simple and predictable controls

- Different communication and interaction preferences

Environmental and Individual Needs

- Sensitivity to temperature or humidity
- Requirement for adjustable lighting or ventilation
- Space for personal accessibility equipment
- Individual preferences for comfort, privacy and sensory stimulation

These considerations will guide the physical layout, controls, electronic interfaces and testing of the prototype.

Emergency, Health and Safety Considerations:

Safety and accessibility will be considered throughout the Aurora Pod's design. As this project is intended to be a student-built prototype rather than a certified medical device, the system will focus on simple, low-risk design choices, clear user controls and reliable basic monitoring.

The pod will not be designed to replace medical equipment, professional care or room-level safety systems.

Ventilation and Environmental Safety

Because the pod uses a partially enclosed fabric structure, maintaining airflow is an important design requirement.

The enclosure will include permanent ventilation openings so that airflow does not depend entirely on powered components. Small low-voltage fans may also improve circulation and reduce heat and humidity buildup.

The system will monitor temperature and humidity electronically and display them to the user. If conditions move outside a user-defined comfort range, the system may provide a visual, audible or vibration-based notification.

Environmental monitoring will primarily inform the user rather than automatically making safety-critical decisions.

Accessible Entry and Emergency Egress

The pod must be easy to enter and leave without requiring fine motor control, high grip strength or complicated fasteners.

The entrance will therefore use a simple opening system that can be operated using a large movement rather than a precise pinch grip. The opening should also be possible from both inside and outside the pod.

No electronic lock or powered mechanism will be required to exit the pod. If power is lost, the user must still be able to leave the space normally.

The entrance and internal layout will provide as much clear space as practical for users with different mobility requirements. Controls and frequently used features will be positioned within comfortable reach from the sleeping area.

Structural Safety

The pod frame will be lightweight and non-load bearing. It will only support the enclosure fabric and lightweight electronic components.

The user's body weight will remain fully supported by the existing bed or mattress structure.

The frame will have a stable base and be positioned so normal interaction with the enclosure does not cause it to fall or shift significantly.

Avoid protruding components, exposed sharp edges, and loose cables wherever possible.

Electrical and Fire Safety

The electronics within the pod will use low-voltage power suitable for student electronics prototyping.

Any mains-powered adapter required for the system will remain outside the main enclosed user space. The prototype will use commercially available power supplies rather than student-built mains-voltage circuitry.

Wiring will be secured away from areas where it could be pulled, trapped or damaged during normal use.

Separate electronic functions may be protected using appropriate current-limiting or circuit-protection methods where necessary.

Avoid components that produce significant heat inside the enclosed space.

Software Reliability

The software will be designed so that a software fault does not prevent the user from leaving the pod or using its basic non-electronic features.

Important functions such as ventilation openings, entry and exit will therefore remain usable without software.

Where appropriate, the microcontroller may include simple error handling and automatic restart behaviour if the program stops responding.

Outputs such as lighting or powered accessories should return to a safe state if the controller loses power.

Included Features

Bed-Level Alignment Monitor

Problem: A mattress, bed frame or adjustable bed can sometimes sit at an unexpected angle. It may also be difficult for some users to judge whether the sleeping surface is level, particularly in low light or when repositioning the bed.

Solution: A small electronic orientation sensor will measure the sleeping surface angle and display a simple level or incline reading. The system may let the user save a preferred position and notify them if the surface moves noticeably away from that setting.

Accessibility: The feature provides an alternative to visually estimating the bed angle. Information can be presented using large text, simple symbols or other feedback methods depending on the final interface.

Users who do not require accessibility support may also use the feature when adjusting pillows, bed supports or their general sleeping position.

Student Feasibility: This feature can be produced using a small orientation sensor connected to the main microcontroller. The software requires basic sensor reading, calibration and threshold comparison, making it suitable for an undergraduate embedded-systems project.

Accessible Entrance Controls

Problem: Small fasteners and narrow handles can be difficult to operate for users with reduced grip strength, limited hand movement, fatigue or visual impairments.

Solution: The main entrance will use large, easy-to-identify handles or pull points that require minimal precision to operate. The entrance mechanism will open with a simple pulling, pushing, or sliding action rather than a small pinch grip.

Accessibility: Controls will be easy to locate by touch and visually distinguishable from the surrounding enclosure. The system should be usable with either hand where practical and should not require significant strength or complex movements. This also makes the entrance easier to locate and operate in low-light conditions.

Student Feasibility: This feature primarily involves physical prototyping and usability testing rather than complex electronics, making it practical to develop alongside the project's electronic elements.

Equipment and Personal-Item Storage

Problem: Loose personal items, electronics and cables around a sleeping area can create clutter and may be difficult to reach from bed.

Solution: The pod will include lightweight storage areas or attachment points positioned within easy reach of the user. These areas could hold small personal items such as a phone, controller, book, water bottle or other frequently used equipment. Storage will be positioned so that items are less likely to fall into the sleeping area.

Accessibility: Storage locations will prioritise reachability and ease of use. Users should not need to stretch far, leave the bed, or perform precise movements to access commonly used items. The layout may be adjusted during user testing to accommodate different reach ranges and preferred sleeping positions.

Student Feasibility: This feature can be developed using simple sewn, clipped or attached storage prototypes and does not require specialist manufacturing.

Accessible Survey and Control Interface

Problem: Using a phone screen at night can be inconvenient for users experiencing fatigue, reduced dexterity, visual difficulty or sensory overload. Small on-screen controls may also make quick interactions more difficult.

Solution: The Aurora Pod will include a simplified physical control interface for common actions without opening a phone application. A large button or small group of clearly separated controls could allow the user to record basic feedback, change settings or respond to prompts. More advanced functions can remain available through the software interface.

Accessibility: Physical controls will use large interaction areas and simple actions. Where practical, the same function may be available through more than one method, such as a physical button and digital interface. This lets users choose the interaction method they find most comfortable. Visual feedback will use simple wording and clear contrast, while avoiding unnecessary flashing lights or bright displays.

Student Feasibility: Push buttons and similar controls can connect directly to the main microcontroller and use straightforward input logic. The initial prototype can therefore start with physical controls before adding more advanced interface features.

Cable Pass-Through Openings

Problem: Users may need to bring charging cables, electronic accessories or other leads into the pod. Routing these through the main entrance can prevent the enclosure from closing properly and may leave cables in inconvenient positions.

Solution: Small cable pass-through openings will be included near suitable edges of the enclosure. The openings will allow cables to enter or leave the pod without keeping the main entrance partially open. They will be designed to avoid sharp bends, trapping, or pulling cables.

Accessibility: Cable openings will be positioned where they can be reached easily from the sleeping area. The user should be able to insert or remove a cable without requiring significant force or precise hand movement. The feature can support chargers, communication devices, accessibility equipment and other personal electronics.

Student Feasibility: The openings can be created as part of the fabric enclosure without requiring complex mechanical parts. Several simple designs can be prototyped and compared during testing.

Temperature and Humidity Monitoring

Problem: A partially enclosed sleeping space may become warmer or more humid than the surrounding room. Users may not immediately notice these changes while resting or sleeping.

Solution: A digital environmental sensor will measure the pod's temperature and humidity. A simple display will show current readings, which may also be stored for later analysis. The user can define preferred

comfort ranges. If readings move outside those ranges, the system may provide a simple notification or increase powered ventilation, where included in the prototype.

Accessibility: The display will prioritise readability through large values, clear labels and strong contrast. The system should not depend entirely on colour to communicate status. Where appropriate, the system may offer different feedback options so users can choose the notification style that works best for them. Environmental preferences will be adjustable, rather than assuming one temperature or humidity range suits every user.

Student Feasibility: Temperature and humidity sensing is well suited to a student embedded-systems project. It requires a sensor, microcontroller and basic display, allowing the system to begin with simple live readings before progressing to data logging, alerts and automatic ventilation control.

Overall Accessibility Approach

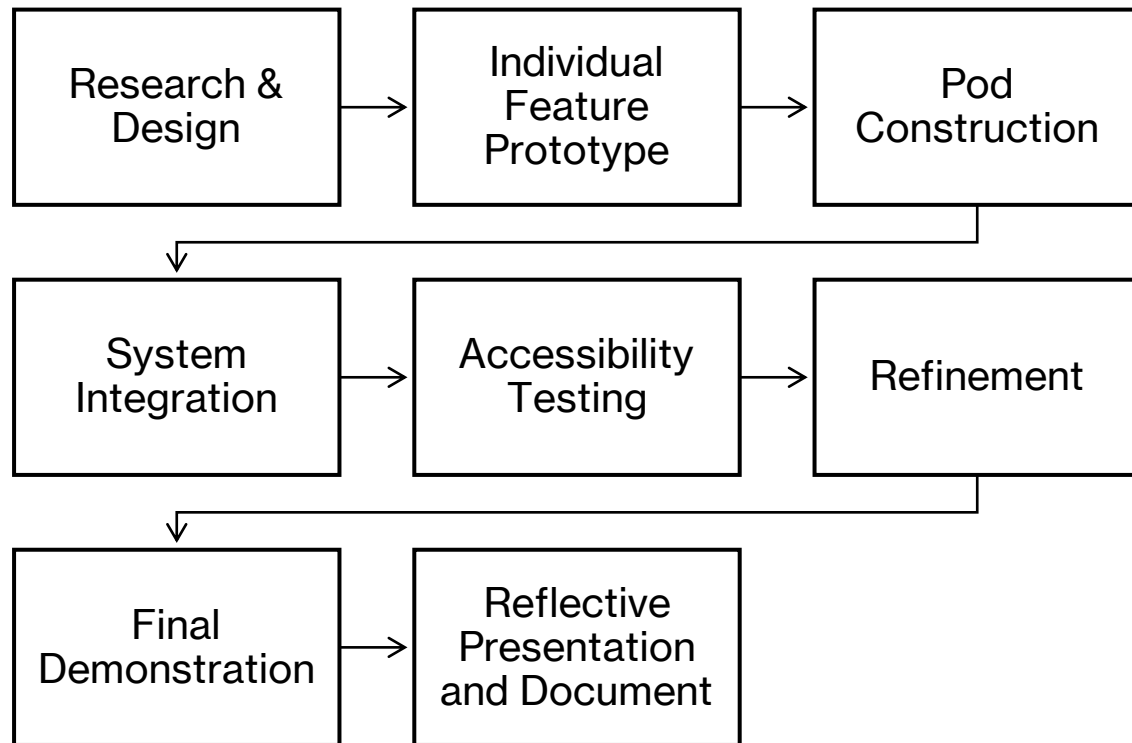
Rather than designing the Aurora Pod around individual medical conditions, the project will use broader inclusive-design principles.

The prototype will aim to:

- minimise the need for fine motor control or high grip strength;
- provide large and easily reached physical controls;
- use clear, high-contrast visual information without relying only on colour;
- reduce unnecessary brightness and sensory stimulation;
- keep important functions usable if electronic systems fail;
- allow environmental and interface preferences to be adjusted by the user;
- keep frequently used controls and storage within comfortable reach;
- provide more than one interaction method where practical;
- avoid unnecessarily complicated opening, locking or control mechanisms; and
- test features with users and revise the design based on usability feedback.

The purpose of the Aurora Pod is therefore not to create a specialist medical enclosure, but to show how embedded electronics, environmental sensing, and inclusive physical design can combine to create a more adaptable and accessible personal space.

Development Plan:



Success Criteria

By the end of the project, the prototype should be functional and demonstrate the accessibility principles established during the design stage. Features will primarily address accessibility needs while remaining useful and intuitive for a wider range of users.

For the prototype to be considered an engineering success, the selected embedded systems should operate as intended individually and when integrated with other features. Relevant sensor readings, testing results and observations will be recorded so that performance can be reviewed and compared throughout development.

Design decisions, testing outcomes and significant technical challenges will also be documented. Features that cannot be successfully implemented within the available time, budget or technical constraints may be redesigned, simplified or excluded from the final prototype with an appropriate engineering justification.

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